

# Data Mining Complete Exam Notes

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · tiers over 17 papers, 2070–2081 BS

## 2. Data Preprocessing

6 hrs · in 17/17 papers · 4 topics, 2 TOP tier · first chapter with numericals → see the Ch 2 Numerical companion.

### Weight of the chapter:

- Data Pre-processing **TOP 12/17** → most-asked topic in the whole syllabus.
- OLAP & Multidim Analysis **TOP 10/17**
- Similarity Measures **HOT 6/17** → nearly always a numerical.
- Data Types & Attributes **PIN 1/17**

### 2.1 Data Types and Attributes

### What is DM? Explain diff data types of attributes in a dataset. 10

**PIN 1/17** 71 Shr DM definition → Ch 1. Only attribute types needed here.

#### Attribute

- Property / characteristic of an object. Eg eye colour, temperature.
- Also called: variable, field, feature, characteristic.
- **Object** = record, point, case, sample, entity, instance.
- Same attr → diff value sets (height in ft or m).
- Diff attrs → same value set, diff properties (ID and age are both int, but ID has no max).

#### Han & Kamber 3rd ed: 4 types (use this, it is the syllabus ref book)

- **Nominal**: names / categories only, no order. Zip code, employee ID, eye colour.
- **Binary**: nominal w/ exactly 2 states (0/1).
  - **Symmetric**: both states equally valuable. Gender.
  - **Asymmetric**: one state rarer + more important. Disease test P/N. **Convention: rarest outcome = 1.**
- **Ordinal**: order meaningful, gap between successive values unknown. {small, medium, large}, grades, mineral hardness.
- **Numeric**: measurable quantity.
  - **Interval-scaled**: gaps meaningful, **no true zero**. Dates, temp in °C/°F.
  - **Ratio-scaled**: gaps and ratios meaningful, true zero. Temp in K, money, counts, age.

#### Older N-O-I-R split (Tan; DBK's slide)

- Nominal, Ordinal, Interval, Ratio. Binary folded into nominal.
- **Same content, different grouping**. Either scores; H&K's is safer as it names binary explicitly, which the similarity questions depend on.

| Attribute Level | Transformation                                                                                              | Comments                                                                                                                               |
|-----------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Nominal         | Any permutation of values                                                                                   | If all employee ID numbers were reassigned, would it make any difference?                                                              |
| Ordinal         | An order preserving change of values, i.e., $new\_value = f(old\_value)$ where $f$ is a monotonic function. | An attribute encompassing the notion of good, better, best can be represented equally well by the values {1, 2, 3} or by {0.5, 1, 10}. |
| Interval        | $new\_value = a * old\_value + b$ where $a$ and $b$ are constants                                           | Thus, the Fahrenheit and Celsius temperature scales differ in terms of where their zero value is and the size of a unit (degree).      |
| Ratio           | $new\_value = a * old\_value$                                                                               | Length can be measured in meters or feet.                                                                                              |

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## 2nd cut: discrete vs continuous

- **Discrete:** finite / countably infinite set. Zip codes, counts, words in a doc. Usually int.
  - **Binary** = special case of discrete.
  - **Symmetric binary:** both states equally valuable (gender).
  - **Asymmetric binary:** one state rarer + more important (disease test P/N). **Matters: Jaccard ignores 0-0 matches, SMC does not.**
- **Continuous:** real values. Temp, height, weight. Stored as float.

## Types of datasets

- **Record**
  - *Data matrix:*  $m$  objects  $\times$   $n$  numeric attrs  $\rightarrow$  points in  $n$ -dim space.
  - *Document data:* each doc = term vector, value = term frequency.
  - *Transaction data:* each record = set of items. Eg grocery basket.
- **Graph:** nodes + links. Eg WWW, molecular structures.
- **Ordered:** spatial (geo coords), temporal ( $d = f(t)$ ), sequential, sequence data.

## Characteristics of structured data

- **Dimensionality**  $\rightarrow$  curse of dimensionality (see 2.2).
- **Sparsity:** % of cells not populated. Sparsity + density = 100.
- **Resolution:** patterns depend on the scale you look at.

## 2.2 Data Pre-processing

### What is Data Pre-Processing? Explain the major tasks performed in it. 2+7 4+4 5+5

**TOP 12/17** 81 Ba, 75 Ch, 72 Ka Most-asked topic in the syllabus.

#### Why preprocess

- Real data is **dirty**: incomplete, noisy, inconsistent.
- Cause: huge size + multiple heterogeneous sources.
- **Garbage in  $\rightarrow$  garbage out.** Low quality data  $\rightarrow$  low quality mining.
- Data prep  $\approx$  **60% of total effort** in KDD.

#### Measures of data quality 73 Shr asks this

- **Accuracy:** values correct.
- **Completeness:** no missing values.
- **Consistency:** no contradictions.
- **Timeliness:** updated in time.
- **Believability:** how far users trust it.
- **Interpretability:** how easily understood.

#### 5 major tasks (**this is the answer**)

1. **Data cleaning:** fill missing, smooth noise, fix inconsistency, rm duplicates.
2. **Data integration:** merge multiple sources into 1 coherent store.
3. **Data transformation:** normalise, aggregate, generalise, construct attrs.
4. **Data reduction:** smaller volume, same analytical result.
5. **Data discretisation:** continuous  $\rightarrow$  intervals (part of reduction).

#### Answer skeleton 7

1. Definition + "why": dirty data, garbage in garbage out.
2. Name the 5 tasks.
3. 2 lines each, with 1 concrete technique per task.
4. Close: cleaning + integration dominate the effort.

## (a) Data cleaning

### Missing values: why

- Equipment malfunction, misunderstanding at entry, deleted as inconsistent, not considered important, no change recorded.

### How to handle missing data 74 Ash, 74 Ch

1. **Ignore the tuple:** usual when class label missing. Bad if % missing varies a lot.
2. **Fill manually:** tedious, infeasible at scale.
3. **Global constant:** eg "Unknown". Risk: mining may treat "Unknown" as a real class.
4. **Attribute mean** of all samples.
5. **Attribute mean of same class** → better than plain mean.
6. **Most probable value:** regression, Bayesian inference, decision tree. **Most popular + most accurate.**

### Noisy data

- Noise = random error / variance in a measured variable.
- Cause: faulty instruments, entry error, transmission error, tech limits, naming inconsistency.

### How to handle noise

- **Binning:** sort, partition into bins, smooth by bin *mean* / *median* / *boundary*.
- **Regression:** fit data to a function.
- **Clustering:** detect + rm outliers.
- **Combined computer + human:** machine flags suspicious, human checks.

### Binning: 2 partitioning rules

- **Equal-width (distance):**  $N$  intervals of equal size.
- **Equal-depth (frequency):**  $N$  intervals, each with  $\approx$  same no. of samples. Better for skewed data.

Equal-width bin width,  $A$  = lowest value,  $B$  = highest,  $N$  = no. of bins:

$$W = \frac{B - A}{N}$$

Your class notes print  $W = (A - B)/N$ , which gives a negative width. Wrong: it is  $(B - A)/N$ . verified/added

## (b) Data integration 5

75 Ash

- Combines data from multiple sources into 1 coherent store.
- **Issues:**
  - **Entity identification:** is `cust_id` in A the same as `cust_no` in B? Solved w/ metadata.
  - **Redundancy:** an attr derivable from another. Detect by **correlation analysis** ( $\chi^2$  for nominal, correlation coefficient + covariance for numeric).
  - **Data value conflicts:** same entity, diff values across sources (diff units, scales, currency, representation).
  - **Duplication** at tuple level.
- Careful integration → reduces/avoids redundancy → improves mining speed + quality.

### $\chi^2$ test of independence (nominal pairs) NEW

- $H_0$ : the 2 attrs are **independent**.  $H_1$ : associated (so 1 is redundant).
- Build the **contingency table** of observed counts  $O_{ij}$ .
- Expected count  $E_{ij} = \frac{(\text{row}_i \text{ total}) \times (\text{col}_j \text{ total})}{\text{grand total}}$
- $\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$ ,  $df = (r - 1)(c - 1)$ .
- $\chi^2 >$  critical value at  $\alpha \rightarrow$  **reject**  $H_0 \rightarrow$  correlated  $\rightarrow$  drop one attr.

### Worked in one line

- Gender (M/F) vs preference (like/dislike), 100 people, all 4 margins = 50.
- Every  $E_{ij} = \frac{50 \times 50}{100} = 25$ . Observed 20/30/30/20.
- $\chi^2 = 4 \times \frac{5^2}{25} = 4$ ,  $df = (2 - 1)(2 - 1) = 1$ .
- Critical value at  $\alpha = 0.05$ ,  $df = 1$  is 3.841.  $4 > 3.841 \rightarrow$  **reject**  $H_0$ , gender and preference are associated.
- **Correlation  $\neq$  causation.** Numeric analogue:  
 $r_{A,B} = \frac{Cov(A,B)}{\sigma_A \sigma_B}$ , formula in §2.4.

**(c) Data transformation + normalisation** 2+6 4 2+2+2

**TOP /17** 73 Shr, 75 Ch, 78 Bh Normalisation numerical is a recurring 6-mark question → full solution in the Numerical companion.

**Transformation methods**

- **Smoothing:** rm noise.
- **Aggregation:** summarise, build data cube.
- **Generalisation:** climb concept hierarchy (street → city → country).
- **Normalisation:** scale into a small range.
- **Attribute construction:** build new attrs from given ones.

**Why normalise**

- Stops large-range attrs dominating small-range ones.
- Required by distance-based methods: KNN, K-means, ANN.

**The 3 normalisation formulas (memorise exactly)**

**1. Min-max** → maps to  $[new\_min, new\_max]$ :

$$v' = \frac{v - min_A}{max_A - min_A} (new\_max_A - new\_min_A) + new\_min_A$$

**2. Z-score** (zero-mean) → use when min/max unknown or outliers present:

$$v' = \frac{v - \bar{A}}{\sigma_A} \quad \bar{A} = \text{mean}, \sigma_A = \text{std deviation}$$

**3. Decimal scaling:**

$$v' = \frac{v}{10^j} \quad j = \text{smallest int s.t. } \max(|v'|) < 1$$

**Variant of 2: z-score w/ mean absolute deviation**  $s_A$ , more robust to outliers than  $\sigma_A$  (deviations are not squared):

$$s_A = \frac{1}{n} \sum_i |v_i - \bar{A}| \quad v' = \frac{v - \bar{A}}{s_A}$$

**Error in your class notes** verified/added : z-score is printed there as  $v' = (v - min)/\sigma$ . That is **wrong**. It is  $(v - \text{mean})/\sigma$ . Using  $min$  makes every  $v' \geq 0$ , which defeats the whole point of a zero-mean score.

**(d) Data reduction** 5

- Smaller volume, **same (or nearly same) analytical result.**
- **Strategies:**
  - **Sampling** (below)
  - **Dimensionality reduction:** PCA, SVD, wavelet
  - **Feature subset selection**
  - **Aggregation** / data cube
  - **Discretisation** + concept hierarchy
  - **Compression:** lossy / lossless
- **Han & Kamber's 3-way split** **NEW** (**safer skeleton for a 5-mark answer**):
  - **Dimensionality reduction:** wavelet transform, PCA, attribute subset selection. *Fewer attributes.*
  - **Numerosity reduction:** replace the data by a smaller representation. *Fewer tuples.*
    - o *Parametric:* fit a model, store only its parameters → regression, log-linear models.
    - o *Non-parametric:* store a reduced form directly → histograms, clustering, sampling, data cube aggregation.
  - **Data compression:** lossless (exact rebuild) vs lossy (approximation). The other two are themselves forms of compression.

### Sampling types

- **Simple random:** equal prob for each item.
- **W/o replacement:** item removed once picked.
- **W/ replacement:** not removed, same item can repeat.
- **Stratified:** split into partitions, sample each. Best for skewed data.

### Feature subset selection

- Rm **redundant** features (purchase price + sales tax paid).
- Rm **irrelevant** features (student ID for predicting GPA).
- **Techniques:**
  - *Brute force:* try all subsets.
  - *Embedded:* selection happens inside the algorithm.
  - *Filter:* select before the algorithm runs.
  - *Wrapper:* use algorithm as a black box to score subsets.

### Discretisation (feeds Ch 3 decision trees)

- Continuous → small no. of interval labels. Many algorithms need discrete input.
- **Unsupervised** (no class labels):
  - Equal-width, equal-frequency, **k-means** (cluster values, 1 bin per cluster).
- **Supervised** (uses class labels): **entropy-based.**
  - $H(p_1, \dots, p_s) = \sum_i p_i \log_2(1/p_i) = - \sum_i p_i \log_2(p_i)$   
**same quantity as *Info(D)* in Ch 3, §3.2** – written both ways in the literature
  - Split  $S$  at boundary  $T$ :  

$$E(S, T) = \frac{|S_1|}{|S|} Ent(S_1) + \frac{|S_2|}{|S|} Ent(S_2)$$
  - Pick  $T$  **minimising  $E$** ; recurse until  

$$Ent(S) - E(T, S) \leq \delta.$$
- Also: **mapping to a new space** (Fourier, wavelet transform).

### Curse of dimensionality 70 Ch 5

- As dims ↑, data becomes increasingly **sparse** in the space it occupies.
- Density + distance lose meaning → breaks clustering + outlier detection.
- All points start looking equidistant.
- **Avoid by:** dimensionality reduction (PCA/SVD), feature subset selection, feature construction, domain knowledge.

**(e) PCA for dimensionality reduction** 10 4

71 Shr 10, 76 Ch 4 numerical Numerical worked in the companion.

**Key idea**

- Find a **projection capturing the largest variance** in the data.
- New axes = eigenvectors of the **covariance matrix**, ordered by eigenvalue.
- Eigenvalue  $\lambda_i$  = variance carried by principal component  $i$ .
- Keep the top  $r$  components  $\rightarrow$  drop the rest w/ little info loss.
- Components are **orthogonal** (uncorrelated).

**Proportion of variance explained**

$$\frac{\sum_{i=1}^r \lambda_i}{\sum_{i=1}^m \lambda_i} = \frac{\lambda_1 + \dots + \lambda_r}{\lambda_1 + \dots + \lambda_m}$$

- Denominator = trace of covariance matrix = total variance.

**PCA process: 5 steps (DBK's sequence)**

1. **Subtract the mean** from each dimension  $\rightarrow$  zero-mean data.
2. **Compute the covariance matrix.**
3. **Compute eigenvalues + eigenvectors** of that matrix.
4. **Choose components, form feature vector:** sort  $\lambda$  descending, keep top  $r$ , put their eigenvectors in columns.
5. **Derive new data:**  $\text{FinalData} = \text{RowFeatureVector} \times \text{RowZeroMeanData}$ .

**Notes**

- Eigenvectors exist only for **square** matrices.
- An  $m \times m$  matrix gives  $m$  of them.
- Eigenvectors of a **symmetric** matrix are mutually perpendicular.
- Normalise eigenvectors to unit length.
- Highly correlated dims  $\rightarrow$  few large  $\lambda \rightarrow$  PCA helps a lot.  
Uncorrelated dims  $\rightarrow r \approx m \rightarrow$  **PCA does not help**.

**Error in your class notes** verified/added : the PCA section there lists “construct a neighbourhood graph” and “compute shortest-path / geodesic distances”. Those steps belong to **Isomap**, a non-linear method, **not** PCA. PCA is purely linear: covariance  $\rightarrow$  eigenvectors.

## 2.3 OLAP & Multidimensional Data Analysis

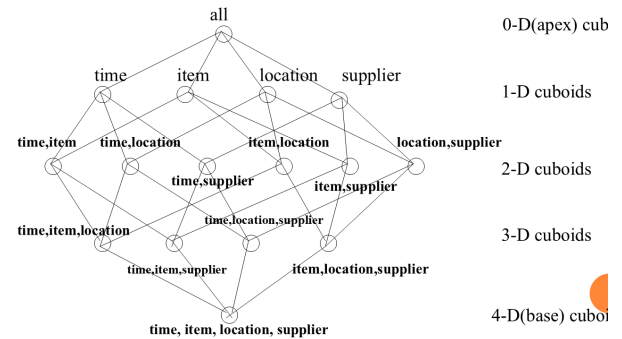
## Explain typical OLAP operations over a multidim DW. Differentiate OLAP vs OLTP. 6+4 10 5+5 2+5+3

TOP 10/17 79 Bh, 72 Ch, 75 Ash, 74 Ch

### Multidimensional model

- **Dimension:** perspective the org keeps records by. Eg time, item, location, branch.
- **Measure:** the numeric quantity analysed. Eg units\_sold, dollars\_sold.
- **Fact table:** stores measures + FKs to dimension tables.
- **Data cube:** lets you view measures against a set of dims.
- **Cuboid:** the cube at one level of summarisation.
  - **Base cuboid** = lowest level, all  $n$  dims.
  - **Apex cuboid** = 0-D, highest summary, denoted all.
  - Lattice of cuboids = the full cube.  $n$  dims  $\rightarrow 2^n$  cuboids.
- Schemas: **star**, **snowflake**, **fact constellation**  $\rightarrow$  defined in Ch 1.
- **Concept hierarchy:** the levels of abstraction inside 1 dim, eg street < city < province < country.
  - This is **what roll-up and drill-down actually climb**.
  - Supplied manually by users/experts, or generated automatically from the data distribution.

### Cube: A Lattice of Cuboids



### 3 kinds of measure, by aggregate function

- **Distributive:** partition the data, aggregate each part, aggregate the results  $\rightarrow$  same answer. `sum()`, `count()`, `min()`, `max()`.
- **Algebraic:** computable from a bounded no. of distributive functions. `avg()` = `sum()/count()`, `min_N()`, `standard_deviation()`.
- **Holistic:** no constant-size sub-aggregate suffices. `median()`, `mode()`, `rank()`. **Expensive**  $\rightarrow$  usually estimated.

### The 5 OLAP operations (the core answer)

- **Roll-up (drill-up):** summarise by climbing a hierarchy or reducing a dim. Eg city  $\rightarrow$  country; or drop time entirely.
- **Drill-down (roll-down):** reverse of roll-up. Summary  $\rightarrow$  detail, or introduce a new dim. Eg quarter  $\rightarrow$  month.
- **Slice:** select on **one** dim  $\rightarrow$  sub-cube. Eg time = Q1.
- **Dice:** select on **two or more** dims. Eg location in {Kathmandu, Pokhara} and time in {Q1,Q2}.
- **Pivot (rotate):** rotate axes to give an alternative presentation. 3-D  $\rightarrow$  series of 2-D planes.
- **Others:** **drill-across** (across  $\geq 2$  fact tables), **drill-through** (down to the back-end relational tables).

### OLAP vs OLTP (asked 4 times)

- **Users:** OLTP clerks / IT · OLAP knowledge workers.
- **Function:** OLTP day-to-day ops · OLAP decision support.
- **DB design:** OLTP application-oriented, normalised (ER) · OLAP subject-oriented, de-normalised (star/snowflake).
- **Data:** OLTP current, detailed, isolated · OLAP historical, summarised, multidim, consolidated.
- **Usage:** OLTP repetitive · OLAP ad-hoc.
- **Access:** OLTP read/write, index on primary key · OLAP lots of scans.
- **Unit of work:** OLTP short simple txn · OLAP complex query.
- **Records accessed:** tens · millions.
- **Size:** 100 MB-GB · 100 GB-TB.
- **Metric:** txn throughput · query throughput / response.

### OLTP vs. OLAP

|                    | OLTP                                                | OLAP                                                              |
|--------------------|-----------------------------------------------------|-------------------------------------------------------------------|
| users              | clerk, IT professional                              | knowledge worker                                                  |
| function           | day to day operations                               | decision support                                                  |
| DB design          | application-oriented                                | subject-oriented                                                  |
| data               | current, up-to-date, detailed, relational, isolated | historical, summarized, multidimensional integrated, consolidated |
| usage              | repetitive                                          | ad-hoc                                                            |
| access             | read/write index/hash on prim. key                  | lots of scans                                                     |
| unit of work       | short, simple transaction                           | complex query                                                     |
| # records accessed | tens                                                | millions                                                          |
| #users             | thousands                                           | hundreds                                                          |
| DB size            | 100MB-GB                                            | 100GB-TB                                                          |
| metric             | transaction throughput                              | query throughput, response                                        |

### OLAP server types (detail in Ch 1, DW architecture)

- **ROLAP:** relational back end. Scales best.

- **MOLAP**: multidim array store. Fastest, wastes space on sparse data.
- **HOLAP**: hybrid. Detail relational + aggregates multidim.

### Base & apex cuboid, slice & dice, roll-up & drill-down w/ example 2+3+3+3

76 Ash Answer = the definitions above + 1 worked example each. Cuboid-count and schema-drawing questions are worked in the Numerical companion  
78 Bh, 76 Ch.

## 2.4 Various Similarity Measures

### How is similarity/dissimilarity calculated? Properties of a distance metric. 3 10 4

**HOT 6/17** 80 Bh, 80 Ba, 78 Bh, 76 Ch, 76 Ash, 71 Ch **Almost always numerical** → companion.

#### Similarity vs dissimilarity

- **Similarity**: how alike 2 objects are. *Higher* = more alike. Usually  $[0, 1]$ .
- **Dissimilarity / distance**: how different. *Lower* = more alike. Min = 0, upper limit varies, often  $[0, \infty]$ .
- Used for: clustering, KNN, outlier detection, search, recommendation, near-duplicate detection.

#### Distance metric: 4 properties (71 Ch asks exactly this, 10 )

- 1. Non-negativity**:  $d(x, y) \geq 0$ .
  - 2. Identity**:  $d(x, y) = 0$  iff  $x = y$ .
  - 3. Symmetry**:  $d(x, y) = d(y, x)$ .
  - 4. Triangle inequality**:  $d(x, y) \leq d(x, z) + d(z, y)$ .
- Triangle inequality → guarantees the function is *well behaved*: the direct connection is the shortest.
  - A measure satisfying all 4 = a **metric**. Minkowski distances are metrics for  $r \geq 1$ .
  - **Subtle point worth 1 mark**: non-negativity is *implied* by the other 3, so strictly only 3 are independent.
  - **Weighted Euclidean** (attrs of unequal importance):  

$$d(i, j) = \sqrt{\sum_k w_k (x_{ik} - x_{jk})^2}$$

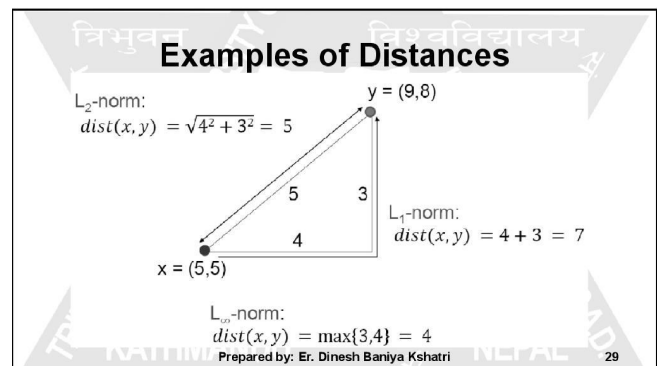
#### Minkowski family: the one formula that generates the rest

$$\text{dist}(p, q) = \left( \sum_{k=1}^n |p_k - q_k|^r \right)^{1/r}$$

- $n$  = no. of dims,  $r$  = parameter,  $r \geq 1$ . **Do not confuse  $r$  with  $n$ .**
- **Notation clash**: H&K call the parameter  $h$  and reserve  $p$  for the no. of attrs; other books write  $L_p$  where their  $p$  is this  $r$ . **Define your symbols.**
- $r = 1 \rightarrow$  **Manhattan** ( $L_1$ , city block, taxicab). Binary case = **Hamming distance**.
- $r = 2 \rightarrow$  **Euclidean** ( $L_2$ ).
- $r \rightarrow \infty \rightarrow$  **Supremum** ( $L_{max}$ ,  $L_\infty$ , Chebyshev, chessboard) = max difference in any single coordinate.

#### Written out

- Euclidean:  $\sqrt{\sum_k (p_k - q_k)^2}$
- Manhattan:  $\sum_k |p_k - q_k|$
- Supremum:  $\max_k |p_k - q_k|$



**Minkowski Distance – Example**

| point | x | y |
|-------|---|---|
| p1    | 0 | 2 |
| p2    | 2 | 0 |
| p3    | 3 | 1 |
| p4    | 5 | 1 |

|    | p1 | p2 | p3 | p4 |
|----|----|----|----|----|
| L1 |    |    |    |    |
| p1 | 0  | 4  | 4  | 6  |
| p2 | 4  | 0  | 2  | 4  |
| p3 | 4  | 2  | 0  | 2  |
| p4 | 6  | 4  | 2  | 0  |

|    | p1    | p2    | p3    | p4    |
|----|-------|-------|-------|-------|
| L2 |       |       |       |       |
| p1 | 0     | 2.828 | 3.162 | 5.099 |
| p2 | 2.828 | 0     | 1.414 | 3.162 |
| p3 | 3.162 | 1.414 | 0     | 2     |
| p4 | 5.099 | 3.162 | 2     | 0     |

|            | p1 | p2 | p3 | p4 |
|------------|----|----|----|----|
| L $\infty$ |    |    |    |    |
| p1         | 0  | 2  | 3  | 5  |
| p2         | 2  | 0  | 1  | 3  |
| p3         | 3  | 1  | 0  | 2  |
| p4         | 5  | 3  | 2  | 0  |

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**Binary attributes: SMC vs Jaccard** 2+2+2

Build a  $2 \times 2$  contingency count for objects  $i, j$ . **Two notations exist, know both:**

- $q = M_{11}$  = attrs where both = 1       $t = M_{00}$  = attrs where both = 0
- $r = M_{10} = i=1, j=0$        $s = M_{01} = i=0, j=1$
- Total  $p = q + r + s + t$ . **H&K use  $q, r, s, t$ ; Tan and DBK use  $M_{11}$  etc.**

$$d = \frac{r+s}{q+r+s+t} \quad d = \frac{r+s}{q+r+s} \quad SMC = \frac{q+t}{q+r+s+t} = 1 - d_{sym}$$

symmetric                      asymmetric

- **Only difference: Jaccard/asymmetric drop  $t$  (the 0-0 matches).**
- **SMC** → symmetric binary (a 0-0 match is informative).
- **Jaccard** → asymmetric binary (0-0 meaningless: 2 people both lacking a rare disease says nothing).
- Sets form:  $J(A, B) = |A \cap B| / |A \cup B|$ .
- **Read the verb.** "Find the Jaccard coefficient" wants  $J$ ; "find the dissimilarity" wants  $d = 1 - J$ .

**Nominal attributes**

$$d(i, j) = \frac{p-m}{p} \quad sim(i, j) = 1 - d(i, j) = \frac{m}{p}$$

- $m$  = no. of attrs where  $i$  and  $j$  are in the **same state**,  $p$  = total attrs.
- Alternative: encode each of the  $M$  states as an **asymmetric binary** attr, then use Jaccard.

**Ordinal attributes**

1. Replace value  $x_{if}$  by its **rank**  $r_{if} \in \{1, \dots, M_f\}$ .
2. Normalise rank to  $[0, 1]$  so every attr has equal weight:

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

3. Treat  $z_{if}$  as **numeric** → use Euclidean / Manhattan.
- Eg {fair, good, excellent},  $M_f = 3 \rightarrow$  ranks 1,2,3 →  $z = 0, 0.5, 1$ .

**Cosine similarity** 3

$$\cos(d_1, d_2) = \frac{d_1 \cdot d_2}{\|d_1\| \|d_2\|}$$

- $d_1 \cdot d_2$  = dot product,  $\|d\| = \sqrt{\sum d_k^2}$  = vector length.
- Range  $[0, 1]$  for non-negative vectors.
- Aligned → angle  $0^\circ \rightarrow \cos = 1$ . Orthogonal (no common terms) →  $90^\circ \rightarrow \cos = 0$ .
- **Ignores magnitude, measures direction only** → ideal for documents, where doc length shouldn't matter.

Which measure for which data (picking wrong = losing the marks)

**SMC vs. Jaccard Example**

$p = 1000000000$   
 $q = 000001001$

$M_{01} = 2$  (the number of attributes where p was 0 and q was 1)  
 $M_{10} = 1$  (the number of attributes where p was 1 and q was 0)  
 $M_{00} = 7$  (the number of attributes where p was 0 and q was 0)  
 $M_{11} = 0$  (the number of attributes where p was 1 and q was 1)

$SMC = (M_{11} + M_{00}) / (M_{01} + M_{10} + M_{11} + M_{00}) = (0+7) / (2+1+0+7) = 0.7$

$J = (M_{11}) / (M_{01} + M_{10} + M_{11}) = 0 / (2+1+0) = 0$

Prepared by: Er. Dinesh Baniya Kshatri 44

**Data matrix vs dissimilarity matrix**

- **Data matrix** ( $n$  objects  $\times$   $p$  attrs): stores the objects themselves. **Two-mode** (rows = objects, cols = attrs).
- **Dissimilarity matrix** ( $n \times n$ ): stores  $d(i, j)$  for every pair. **One-mode** (both axes are objects).
- Properties:  $d(i, i) = 0$ ,  $d(i, j) = d(j, i) \rightarrow$  only the **lower triangle** is written.
- Most clustering + KNN algorithms run on the dissimilarity matrix.

**Mixed attribute types**

$$d(i, j) = \frac{\sum_{f=1}^p \delta_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^p \delta_{ij}^{(f)}}$$

- $\delta_{ij}^{(f)} = 0$  if the attr is missing for either object, or it is asymmetric binary and both are 0. Else  $\delta = 1$ .
- $d_{ij}^{(f)}$  computed per attr type, each scaled to  $[0, 1]$  (numeric: divide by the range).

**Correlation coefficient**

$$corr(x, y) = \frac{\sum_i (x_i - \mu_x)(y_i - \mu_y)}{\sqrt{\sum_i (x_i - \mu_x)^2} \sqrt{\sum_i (y_i - \mu_y)^2}} = \frac{s_{xy}}{s_x s_y}$$

- = cosine similarity of the **mean-centred** vectors.
- Range  $[-1, 1]$ : +1 positive, -1 negative, 0 no correlation.
- **Covariance**  $s_{xy} = \frac{1}{n-1} \sum_k (x_k - \bar{x})(y_k - \bar{y})$ .
- Correlation = scaled (unit-free) version of covariance.

| Data                                        | Use                                                                       |
|---------------------------------------------|---------------------------------------------------------------------------|
| Numeric, scale matters                      | Euclidean / Manhattan / Minkowski                                         |
| Numeric, only direction matters (documents) | Cosine                                                                    |
| Symmetric binary                            | SMC                                                                       |
| Asymmetric binary, sets                     | Jaccard                                                                   |
| Nominal / categorical                       | Simple mismatch $d = (p - m)/p$ , $m$ = no. of matches, $p$ = total attrs |
| Strings                                     | Hamming (equal length) / Levenshtein edit distance                        |
| Testing linear relationship                 | Correlation coefficient                                                   |

## 2.5 Basic Statistical Descriptions of Data

**NEW** Han & Kamber 3rd ed §2.2. **Never asked in the 17 papers**, so no tier chip. Added because PC's 2025–26 Unit 2 handout and its numerical companion are built entirely on it, and because the IQR outlier rule below is the statistical basis of the boxplot test used in Ch 6. **Learn the two boxed formulas; skim the rest.**

## Measures of central tendency

- **Mean**  $\bar{x} = \frac{1}{n} \sum_i x_i$  (sample),  $\mu = \frac{1}{N} \sum x$  (population).
- **Weighted mean**  $\bar{x} = \frac{\sum w_i x_i}{\sum w_i}$ .
- **Trimmed mean**: chop the extreme values first → kills outlier sensitivity. Chop at most 2% per end.
- **Median**: middle value if  $n$  odd, mean of the middle two if  $n$  even.
- **Median is holistic** → needs a full sort → expensive on a large table. (Same holistic/distributive/algebraic split as the cube measures, §2.3.)
- **Mode**: most frequent value. Unimodal / bimodal / trimodal / multimodal. No mode if every value occurs once.
- **Midrange** =  $\frac{\max + \min}{2}$ . Cheap: 1 pass,  $\max()$  and  $\min()$ .

**Median of grouped data** (intervals, not raw values):

$$\text{median} = L_1 + \left( \frac{N/2 - (\sum \text{freq})_l}{\text{freq}_{\text{median}}} \right) \times \text{width}$$

$L_1$  = lower boundary of the median interval,  $N$  = total count,  $(\sum \text{freq})_l$  = cumulative frequency of all intervals below it,  $\text{freq}_{\text{median}}$  = its own frequency,  $\text{width}$  = its width.

**Locate the interval first**: the one where cumulative frequency first reaches  $N/2$ .

**Empirical relation**, unimodal and moderately skewed:

$$\text{mean} - \text{mode} \approx 3(\text{mean} - \text{median})$$

$$\Rightarrow \text{mode} \approx 3\text{median} - 2\text{mean}$$

## Skew, read off mean vs median vs mode

- **Symmetric**:  $\text{mean} = \text{median} = \text{mode}$ .
- **Positively skewed** (right tail):  $\text{mode} < \text{median} < \text{mean}$ .
- **Negatively skewed** (left tail):  $\text{mean} < \text{median} < \text{mode}$ .
- **Mean is dragged toward the tail**, median is not. That is why median is the robust one.

## Measures of dispersion

- **Range** =  $\max - \min$ .
- **Quartiles**:  $Q_1$  = 25th percentile,  $Q_2$  = median,  $Q_3$  = 75th percentile.
- **IQR** =  $Q_3 - Q_1$  = spread of the middle half.
- **Variance**  $s^2 = \frac{1}{n-1} \sum_i (x_i - \bar{x})^2$ , **std deviation**  $s = \sqrt{s^2}$ .
- Population forms use  $\frac{1}{N}$  and  $\sigma$ .

## Two quartile conventions (they disagree, say which you used)

- **Positional (Han & Kamber, the syllabus book)**: for  $n = 12$  take the 3rd, 6th and 9th sorted values directly.
- **Interpolated (PC's class notes)**: position of  $Q_k = \frac{k(n+1)}{4}$ , then linear interpolation between the two neighbouring values.
- Both are marked correct. **The outlier verdict comes out the same either way** → see Ch 2 Numerical, M7.

## Five-number summary and boxplot

- **Five-number summary**, in this order: *Minimum,  $Q_1$ , Median,  $Q_3$ , Maximum*.
- **Boxplot**: box spans  $Q_1$  to  $Q_3$  (box length = IQR), line inside marks the median, two **whiskers** extend to min and max, outliers plotted as individual points.

**IQR outlier rule (the one to memorise)**: a value is a suspected outlier if

$$x < Q_1 - 1.5 \text{ IQR} \quad \text{or} \quad x > Q_3 + 1.5 \text{ IQR}$$

Beyond  $3 \text{ IQR}$  = **extreme outlier**. **Used again in Ch 6** as the boxplot test.

## Graphic displays

- **Histogram**:  $x$  = value / bin,  $y$  = frequency.
- **Quantile plot**: each  $x_i$  against  $f_i$ , the fraction of data at or below it.
- **Quantile-quantile (q-q) plot**: quantiles of one distribution against the corresponding quantiles of another → compares 2 datasets.
- **Scatter plot**: each pair as a point → shows correlation and clusters.
- **Histogram often tells more than a boxplot**: two datasets can share a boxplot and have completely different histograms.

## 2.6 PYQ Mapping

Tiers: **TOP /17** = 7+/17 **HOT /17** = 4-6 **PIN /17** = 1-3 but  $\geq 8$  marks. **N** marks a question fully solved in the Ch 2 Numerical companion.

## 2.1 Data Types and Attributes

1. **PIN 1/17** "What is data mining? Explain different data types of attributes in a dataset."

## 2.2 Data Pre-processing

### 2. TOP 12/17 Why preprocess + major tasks / methods.

- “What is Data Pre-Processing? Briefly explain the major tasks performed.” 2+7 81 Ba
- “Why data preprocessing is necessary? Explain methods to maintain data quality.” 4+4 75 Ch
- “Why is it required? Explain approaches of data cleaning.” 5+5 72 Ka
- 3. PIN 1/17 “Measuring elements of data Quality? Data transformation by normalization w/ example.” 2+6 73 Shr
- 4. N “Use the following methods to normalize 200, 300, 400, 600, 1000” (min-max, z-score, decimal scaling). 2+2+2 78 Bh
- 5. PIN 2/17 Handling missing values. 5 74 Ash 2 74 Ch
- 6. PIN 1/17 Issues in Data Integration. 5 75 Ash
- 7. PIN 1/17 “What is ‘curse of Dimensionality’? How can it be avoided?” 5 70 Ch paper typo: “curse”
- 8. PIN 1/17 “Discuss the impact of noisy data in data mining.” 5 70 Ch
- 9. PIN 1/17 “How can PCA be used for dimensionality reduction?” 10 71 Shr
- 10. N “Find the principal components + proportion of total variance” ( $3 \times 3$  covariance matrix). 4 76 Ch
- 11. PIN 3/17 Short notes: data smoothing 76 Ash · data normalization 75 Ch · data transformation 73 Shr, 72 Ka. 3 4

## 2.3 OLAP & Multidimensional Data Analysis

### 12. TOP 10/17 OLAP operations + OLAP vs OLTP.

- “Explain typical OLAP operations...Differentiate OLAP and OLTP tools.” 6+4 79 Bh
- “How do you perform analysis of multidimensional data? Explain w/ OLAP.” 10 72 Ch
- “Discuss Data Integration issues. Describe OLAP + operations w/ example.” 5+5 75 Ash
- “Approaches to handle missing data? OLAP + operations? OLAP vs OLTP.” 2+5+3 74 Ch
- 13. PIN 1/17 “Dimensional data? Base & apex cuboid? Slicing & Dicing? Roll Down/Up? Give example.” 2+3+3+3 76 Ash
- 14. N Star schema for a 4-dim DW + which OLAP ops list total charge. 3+3 78 Bh
- 15. N Snowflake schema for a 5-dim DW + OLAP ops for count per state per year. 3+3 76 Ch
- 16. PIN 3/17 Short notes: OLAP Operations 74 Ash, 73 Shr · OLAP and OLTP 75 Ch. 4

## 2.4 Various Similarity Measures

- 17. N Cosine sim (Obj 2,4) + Euclidean (1,3) and (1,4), 4-attribute object table. 3+3+3 80 Bh
- 18. N Jaccard (asymmetric) + symmetric-binary dissimilarity + SMC + cosine of 2 doc vectors. 2+2+2+1 80 Ba
- 19. N Manhattan + Supremum distance matrix · cosine of 2 vectors · Jaccard + SMC of 2 binary vectors. 2+1+2 76 Ch
- 20. PIN 1/17 “Properties a Distance Metric needs to support w.r.t. Minkowski’s distance.” 10 71 Ch
- 21. PIN 1/17 “List types of similarity measures + their application areas.” 3 80 Ba
- 22. PIN 2/17 Short notes: Minkowski Distance 78 Bh · similarity measures between tuples 76 Ash. 4 3

### Coverage check

- All Ch 2 PYQs covered. 8 are numerical → fully solved in the companion.
- **3 errors in your source notes corrected here** verified/added :
  - Z-score written as  $(v - \min)/\sigma \rightarrow$  must be  $(v - \text{mean})/\sigma$ .
  - Equal-width bin width  $W = (A - B)/N \rightarrow$  must be  $(B - A)/N$ .
  - PCA section contaminated w/ Isomap steps (neighbourhood graph, geodesic distances).
- **Gap filled by research** verified/added : similarity-measure formulas were absent from the typed notes (they say only “Refer Class Note”), so they were taken from DBK Lecture-2 and then **verified against Han & Kamber 3rd ed** (the syllabus reference book).

### Sources used for this chapter

- **Han & Kamber, 3rd ed** (full textbook PDF): ch 2 attribute types + all proximity measures, ch 3 preprocessing + normalisation, ch 4 data cube + OLAP. **Primary authority.**
- **DBK Lectures 1–4** (Thapathali class slides): attribute types, binning example, discretisation, distance/similarity, PCA process, data cube + concept hierarchies.
- data\_mining\_/Chapter\_1\_2\_.pdf (Bhupesh K. Mishra), Chapter3\_0.ppt, Chapter\_2\_Data Warehouse.ppt.
- **Notes by PC** (Er. Prakash Chandra Prasad, Asst. Prof., IOE Pulchowk), dated 2025–26: Unit 2.docx, unit 2 Numerical.docx, Redundancy and Correlation Analysis.docx. Source of the new §2.5 and of the  $\chi^2$  procedure in §2.2(b). **Newest teaching material available**, so treat §2.5 as insurance even though no past paper has asked it.
- Where sources disagree, **Han & Kamber 3rd ed wins** and the clash is flagged inline.

## Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

| Short    | Means                            | Short   | Means                         |
|----------|----------------------------------|---------|-------------------------------|
| DM       | data mining                      | DW      | data warehouse                |
| DB       | database                         | DBMS    | database management system    |
| KDD      | knowledge discovery in databases | DSS     | decision support system       |
| OLTP     | on-line transaction processing   | OLAP    | on-line analytical processing |
| ROLAP    | relational OLAP                  | MOLAP   | multidimensional OLAP         |
| HOLAP    | hybrid OLAP                      | ETL     | extract, transform, load      |
| BI       | business intelligence            | ER      | entity–relationship           |
| attr(s)  | attribute(s)                     | dim(s)  | dimension(s)                  |
| multidim | multidimensional                 | txn(s)  | transaction(s)                |
| info     | information                      | ops     | operations                    |
| repo     | repository                       | freq    | frequent / frequency          |
| config   | configuration                    | no.     | number                        |
| PCA      | principal component analysis     | SVD     | singular value decomposition  |
| SMC      | simple matching coefficient      | KNN     | <i>k</i> -nearest neighbour   |
| ANN      | artificial neural network        | POS     | point of sale                 |
| FK       | foreign key                      | UI      | user interface                |
| GIGO     | garbage in, garbage out          | w/, w/o | with, without                 |
| rm       | remove                           | vs      | versus                        |
| Eg       | example / for example            | +Fig    | draw a figure                 |
| →        | leads to / therefore             | +       | and                           |

**Exam-paper month codes** (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

**Tier chips:** **TOP /17** = topic asked in 7 or more of the 17 papers · **HOT /17** = 4–6 papers · **PIN /17** = 1–3 papers but worth  $\geq 8$  marks. **Mark chips** such as **2+5** are the marks that question actually carried. **verified/added** marks anything researched and added that was not in the source notes.